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Recent research on *Spirulina* in Italy

L. Tomaselli, G. Torzillo, L. Giovannetti, B. Pushparaj, F. Bocci, M. Tredici, T. Papuzzo, W. Balloni & R. Materassi
Centro di Studio dei Microrganismi Autotrofi del CNR ed Istituto di Microbiologia Agraria e Tecnica, Università di Firenze, Piazzale delle Cascine 27, 50144 Firenze, Italy

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Introduction

As with other microbiological processes carried out on an industrial scale, the successful exploitation of *Spirulina* requires the availability of improved strains and an adequate development of the biotechnology for outdoor mass culture and for processing the biomass. Work in progress at the Research Centre on Autotrophic Microorganisms at Firenze is directed both to better understanding the physiological properties of *Spirulina platensis* and *S. maxima* (syn. *S. geitleri*) and to devising different culture techniques suitable for application on an industrial scale under different environmental situations.

In this report some data on the growth properties of a number of strains of *S. platensis* and *S. maxima* are reported. Also, the results of an investigation on the growth of *Spirulina* in fertilized seawater in laboratory and outdoor cultures are presented.

Characterization of *Spirulina* strains

A comparative investigation on the growth properties of ten strains of *Spirulina platensis* and eight strains of *S. maxima* has been undertaken. Three strains of *S. platensis* were obtained from other laboratories (strain C1 from Leonard, strain LB 1475/4 from the Cambridge Culture Collection and strain M135 from Teronobu's collection). The remaining 15 strains have been isolated in our laboratory, mostly from water samples collected from Lake Texcoco (Mexico) and lakes Monbolo and Rombou (Chad). The properties taken into consideration include: relative photosynthetic efficiency under photolimited conditions, protein production, rate of dark respiration, behavior at temperatures above the optimum for growth, and tolerance of salinity.

The relative photosynthetic efficiency under photolimited conditions was evaluated using Roux bottles of 1-L capacity incubated in a water bath at 30 °C and irradiated intermittently from one side with a battery of fluorescent lamps (PAR intensity equal to 65.3 $J \cdot m^{-2} \cdot s^{-1}$). The light:dark cycle was 12:12 h. The thickness of the culture vessel was 4 cm. The initial biomass concentration was 350 mg (dry wt)$\cdot L^{-1}$. The cultures were run for six complete cycles. The data reported refer to the third and fourth cycles. The culture solution was the standard bicarbonate-carbonate medium (Paoletti *et al.*, 1975). The pH was maintained in the range of 9–9.5 by bubbling air + 1% CO_2.

Table 1 shows a summary of data relative to the amount of biomass produced in the 12-h light period and to the amount of biomass lost through respiration during the 12-h dark period. Large differences among different strains were observed for both species. In contrast, no significant differences between the two species in the amount of biomass synthesized during the light period were observed. On average, the yield of biomass achieved by *S. platensis* during the light period was slightly (about 5 %) higher than the yield of *S. maxima*. An even greater variability was observed in the rate of

Table 1. Biomass production during a 12-h light period (A) and amount of biomass lost through respiration during the 12-h dark period (B) by ten strains of *S. platensis* and eight strains of *S. maxima* (data in $mg \cdot L^{-1}$).

	S. platensis		*S. maxima*	
	A	B	A	B
Maximum	346	127	329	118
Minimum	193	27	167	13
Mean	266	82	251	42

dark respiration in both species. The amount of biomass lost during the dark, expressed as per cent of the biomass produced during the light period, ranged from 14–40 % in *S. platensis* and from 7–35 % in *S. maxima. On average, S. platensis* showed a much higher dark-respiration rate than *S. maxima* (30.5% loss of biomass in the former *vs* 15.7% in the latter). Consequently, the net productivity for a complete light:dark cycle was higher in *S. maxima* (209 $mg \cdot L^{-1} \cdot d^{-1}$) than in *S. platensis* (185 $mg \cdot L^{-1} \cdot d^{-1}$). The dark-respiration rate has been measured at 30 °C and it would be interesting to verify the behavior at temperatures in the range of 15–20 °C, typical of night temperatures in outdoor culture.

In Table 2 the time course of the biosynthesis and consumption of biomass, carbohydrates and proteins during a complete light:dark cycle is shown. In *S. platensis* during the light period, more carbohydrates than proteins are produced. During the night a considerable amount of protein is formed at the expense of carbohydrates. As a result the composition of the biomass in *S. platensis* varies appreciably during a light:dark cycle. In *S. maxima*, proteins are produced mostly during the light period and the variations in biomass composition are less pronounced.

Table 2. Changes in the amount of total biomass, carbohydrates and proteins during a complete light-dark cycle (data in $mg \cdot L^{-1}$ of culture).

	Spirulina platensis M2			*Spirulina maxima* 3Mx		
	A[1]	B	C	A	B	C
Total biomass	680	980	830	630	880	815
Proteins	420	480	520	400	490	510
Carbohydrates	110	270	130	90	180	140

[1]A = Start of light period; B = end of light period; C = end of dark period.

Within both species marked differences in the maximum temperature for growth have been found. While some strains are unable to grow above 36 °C, others can grow above 40 °C. A strain of *S. platensis* able to grow up to 42 °C has been successfully grown in a tubular photobioreactor. When grown at temperature close to the upper limit, all the strains tested show a drastic reduction in protein content, and a corresponding increase in carbohydrate content. These changes are more pronounced as the incubation temperature increases.

Cultivation of *Spirulina* in seawater

The development of a procedure for mass-culturing *Spirulina* in seawater is recommended for at least two reasons. Firstly, vast regions of the world with favorable climatic conditions for growing *Spirulina* are characterized by an acute deficiency of fresh water, while seawater is available in unlimited amounts in coastal areas. On the other hand, the use of seawater reduces the consumption of chemicals for the formulation of the culture medium.

Previous approaches to growing *Spirulina* in seawater were based on a preliminary chemical treatment to remove calcium and magnesium because it was claimed that too high a content of these cations is detrimental to *Spirulina*. Moreover, at the high pH required by this organism, the addition of phosphates and carbonates may produce insoluble Ca and Mg salts. Of course the additional cost for preliminary treatment of the seawater made its utilization less attractive. A few years ago we achieved the culture of *Spirulina* in untreated and fertilized seawater under laboratory conditions (Materassi *et al.*, 1984). The basic requirements for avoiding the formation of insoluble phosphates and carbonates are a close control of pH near 8.5, obtained through addition of CO_2 regulated by a 'pH-stat' system; a controlled supply of phosphate in order to maintain the concentration of PO_4^{3-} below 10 mg $\cdot L^{-1}$; and an initial addition of HCO_3^- not exceeding 1 $g \cdot L^{-1}$.

This procedure has been tested in outdoor culture in 1984 and 1985. The experiments were carried out in southern Italy (Calabria), where the climatic conditions allow *Spirulina* to grow outdoors during the whole year (Tredici *et al.*, 1986). In Fig. 1 the yield of biomass obtained in the one-year experiment in Calabria is reported. The control medium was the standard bicarbonate medium utilized in our Research Centre for the mass culture of *Spirulina*. Nitrate was the nitrogen source in the control medium, while nitrate and urea were tested in the seawater medium. The mean daily yields of biomass (including also the winter season) were 8.1 g (dry wt) $\cdot m^{-2}$ in the control medium, 7.3 $g \cdot m^{-2}$ in seawater plus urea, and 5.2 $g \cdot m^{-2}$ in seawater plus nitrate. The difference between the two nitrogen sources seems too large to be accounted for solely by the lower growth yield due to the additional demand of reducing equivalents on nitrate.

Table 3. Crude protein (N × 6.25) content (%) of *Spirulina maxima* grown in outdoor culture in seawater and in bicarbonate medium.

	Seawater + KNO_3	Seawater + urea	Control medium (nitrate)
Summer	56	61	64
Winter	32	52	43
Mean of one year	44	53	56

The nitrogen content of the biomass grown in seawater plus nitrate is much lower than that of the biomass grown in the control medium. No great differences between control medium and seawater plus urea were observed. Some impairment in the uptake of nitrate or its metabolism seems to occur in seawater. Evidence supporting this hypothesis has been obtained by studying the effects of a sud-

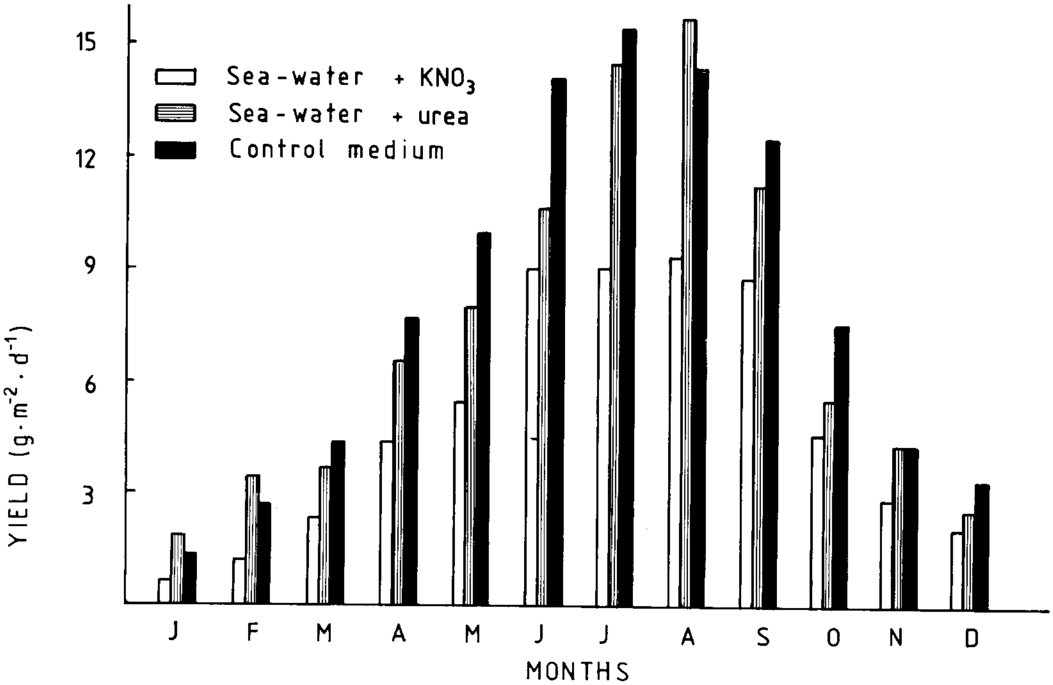

Fig. 1. Yield of biomass by outdoor cultures of *Spirulina maxima* grown in seawater and in the standard bicarbonate medium.

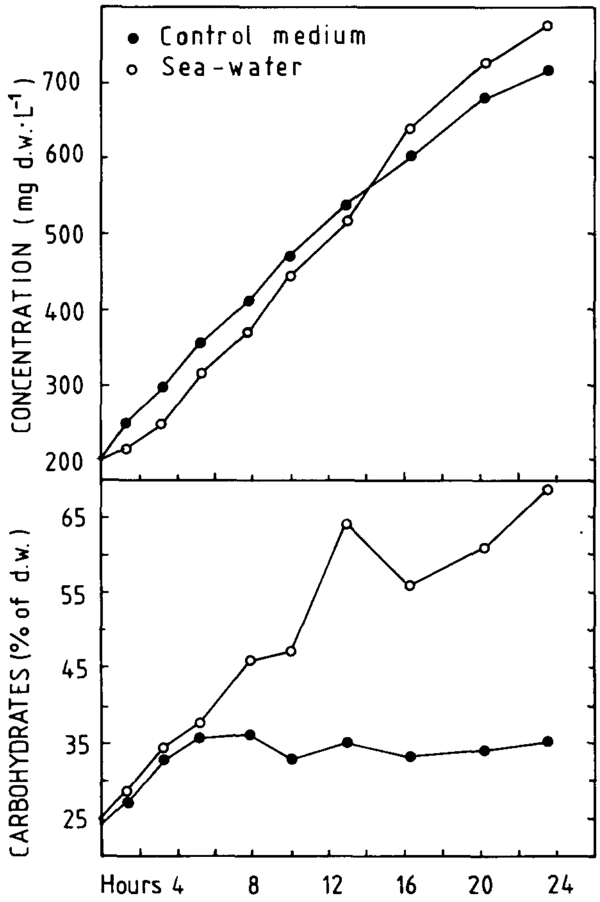

Fig. 2. Effect of a 'light shock' on growth and carbohydrate synthesis by *Spirulina maxima* grown at 30 °C under artificial irradiation (200 $\mu E \cdot m^{-2} \cdot s^{-1}$) in seawater and in the bicarbonate medium. The 'light shock' was obtained at time 0 by diluting to a biomass concentration of 150 mg (dry wt)$\cdot L^{-1}$. Cultures were maintained for ten generations at a cell density of 800–1000 mg$\cdot L^{-1}$.

den increase in light intensity on the biosynthesis of proteins by *S. maxima* (Fig. 2).

Conclusions

The considerable differences in the growth properties of ten strains of *S. platensis* and eight strains of *S. maxima* justify the opinion that there is considerable potential for strain improvement through selection.

The outdoor mass culture of *Spirulina* in seawater is technically feasible without any pretreatment. However, several physiological aspects of the growth of *Spirulina* in seawater remain to be clarified in order to derive the best benefits from this opportunity.

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